OXFORD

INTERNATIONAL AQA EXAMINATIONS

Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname ______________________

Forename(s) ______________________

Candidate signature ______________________

I declare this is my own work.

INTERNATIONAL GCSE

BIOLOGY

Paper 1

Wednesday 26 October 2022 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

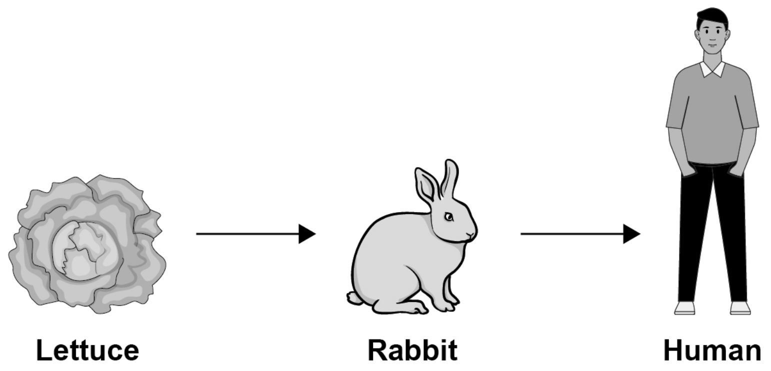
For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

Answer **all** questions in the spaces provided.

0 1 This question is about biomass and energy.

Figure 1 shows a food chain.

Figure 1



0 1 . 1 What energy does the lettuce absorb to make food molecules?

[1 mark]

Tick (✓) **one** box.

Heat ☐

Light ☐

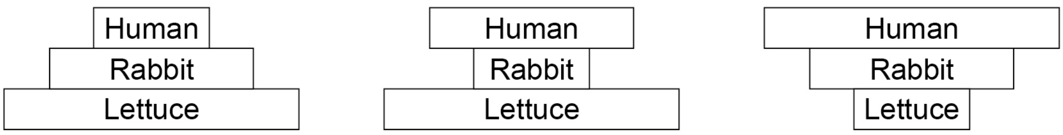
Sound ☐

0 1 . 2 The biomass of each stage of a food chain can be represented in a pyramid of biomass.

Which pyramid of biomass represents the food chain in **Figure 1**?

[1 mark]

Tick (✓) **one** box.



☐ ☐ ☐

Do not write outside the box

0 1 . 3 Energy can be measured in joules (J).

174 000 J of energy reach the lettuce in 1 day.

The lettuce absorbs 1914 J of energy.

Calculate the percentage of energy that the lettuce absorbs.

[2 marks]

____

____

Percentage = ________________ %

0 1 . 4 Chloroplasts in plant cells absorb energy.

Plant cells contain several other parts.

Draw **one** line from each cell part to its function.

[3 marks]

Cell part	**Function of the part**
	Contains cell sap
Mitochondria	
	Controls the activity of the cell
Nucleus	
	Controls what enters and leaves the cell
Vacuole	
	Releases energy for the cell

Question 1 continues on the next page

Turn over ►

Figure 1 is repeated below.

Figure 1

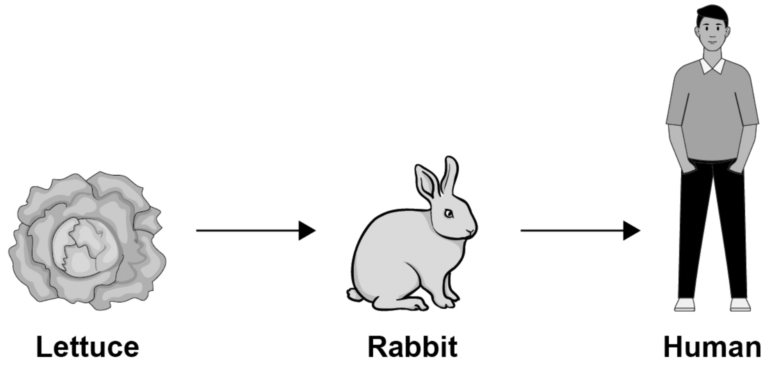
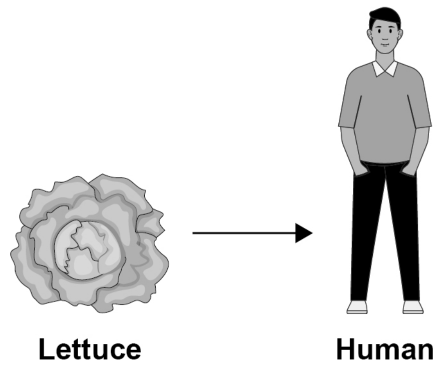


Figure 2 shows a different food chain.

Figure 2



0 1 . 5 The food chain in **Figure 2** is more efficient than the food chain in **Figure 1** in making food for humans.

Explain why the food chain in **Figure 2** is more efficient.

[2 marks]

Do not write outside the box

0 1 . 6 The lettuces that are not eaten die and decay.

Complete the sentences.

[2 marks]

The dead lettuces decay in a process called ____________________________.

The process is carried out by ____________________________.

0 1 . 7 Substances are released when lettuces decay.

The substances help new plants grow.

Which **two** substances are released?

[2 marks]

Tick (✓) **two** boxes.

Amylase	☐
Carbon dioxide	☐
Nitrate ions	☐
Oxygen	☐
Starch	☐

13

Turn over for the next question

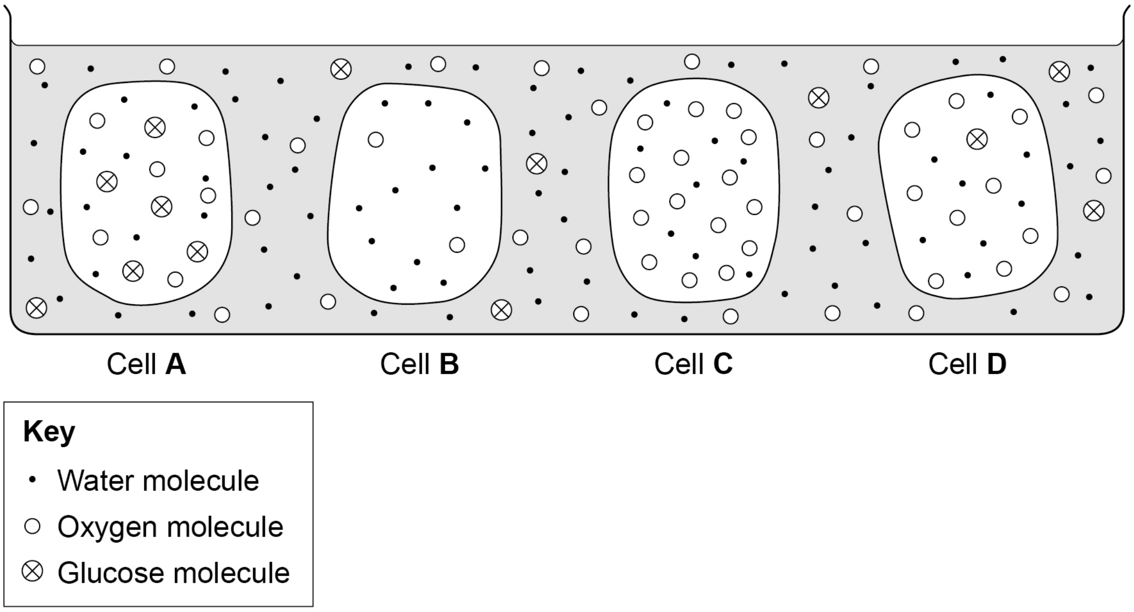
Do not write outside the box

0 2 Substances can be transported into and out of cells by:

- diffusion
- osmosis
- active transport.

Figure 3 shows four cells in a solution.

Figure 3



0 2 . 1 Which cell will oxygen diffuse into?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐ **D** ☐

0 2 . 2 Which chemical reaction in cells requires oxygen?

[1 mark]

____

0 2 . 3 Which cell will use active transport to absorb glucose molecules?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐ **D** ☐

0 2 . 4 What is needed to move glucose molecules into a cell by active transport?

[1 mark]

Tick (✓) **one** box.

Carbon dioxide ☐

Energy ☐

Ions ☐

A student investigated osmosis in potato cells.

This is the method used.

1. Prepare a sugar solution at a concentration of 1.0 mol/dm^3.
2. Cut six potato pieces of equal size.
3. Put 50 cm^3 of the 1.0 mol/dm^3 sugar solution into a labelled boiling tube.
4. Record the mass of a potato piece and place in the boiling tube.
5. Repeat steps 3 and 4 for five other concentrations of sugar solution.
6. After 1 hour remove each potato piece and gently dry with a paper towel.
7. Record the mass of each potato piece.
8. Calculate the percentage change in mass for each potato piece.

0 2 . 5 Describe how to **accurately** prepare 50 cm^3 of 0.2 mol/dm^3 sugar solution using distilled water and the 1.0 mol/dm^3 sugar solution.

You should include the equipment you would use.

[2 marks]

____

____

____

____

0 2 . 6 What is the independent variable in this investigation?

[1 mark]

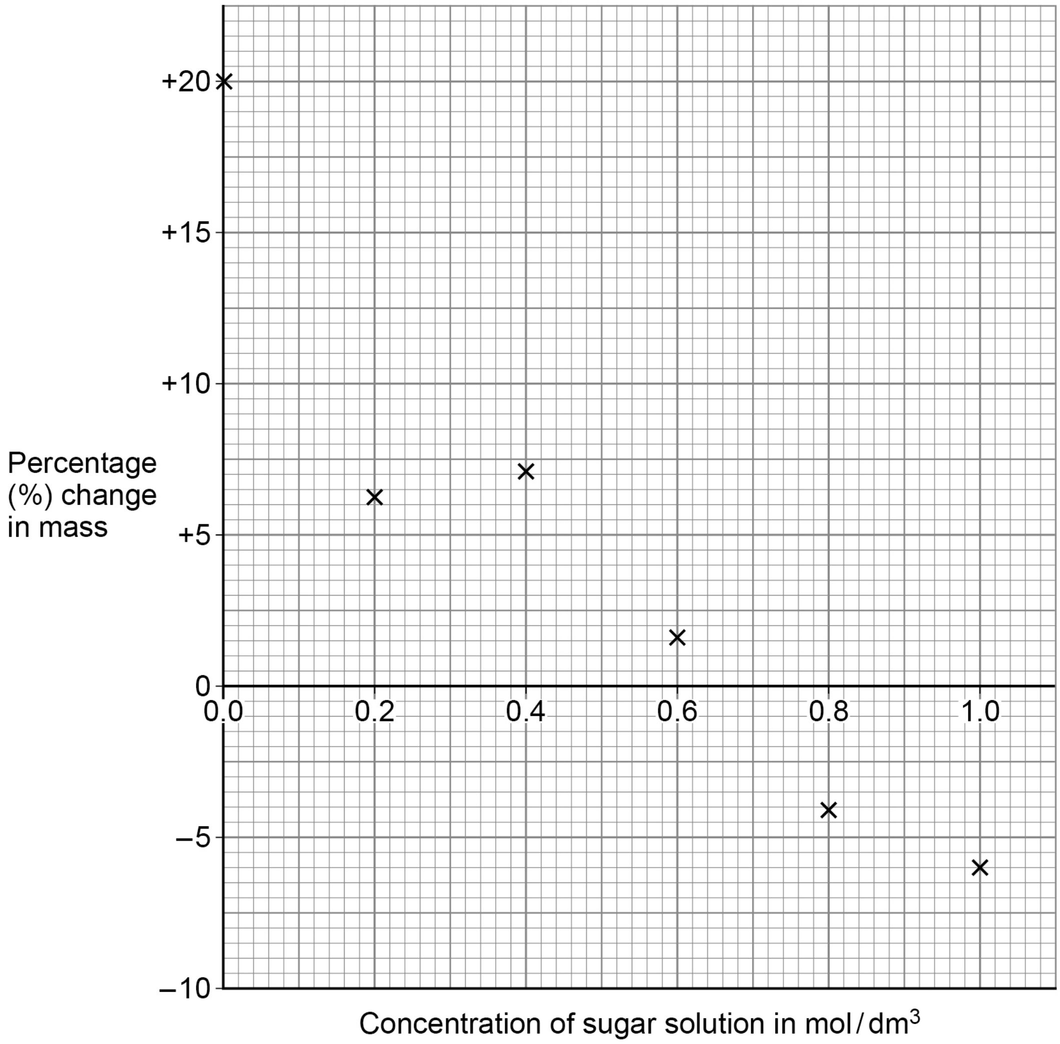
____

Table 1 and **Figure 4** show the results.

Table 1

Concentration of sugar solution in mol/dm^3	Starting mass of potato piece in grams	End mass of potato piece in grams	Percentage change in mass
0.0	1.20	1.44	+20.0
0.2	1.10	1.17	+6.4
0.4	1.12	1.20	+7.1
0.6	1.25	1.27	+1.6
0.8	1.23	1.18	–4.1
1.0	1.17	1.10	–6.0

Figure 4



Do not write outside the box

0 2 . 7 The result at 0.2 mol/dm^3 sugar solution is anomalous.

What evidence in **Figure 4** shows the result is anomalous?

[1 mark]

0 2 . 8 Suggest an error that may have caused the anomalous result.

[1 mark]

0 2 . 9 Determine the concentration inside the potato cells.

You should draw a line of best fit on **Figure 4**.

[2 marks]

Concentration = ____________________ mol/dm^3

11

Turn over for the next question

Do not write outside the box

0 3 This question is about cells.

0 3 . 1 Name the type of cell division that produces new skin cells.

[1 mark]

____

0 3 . 2 Give **two** reasons why new skin cells are needed.

[2 marks]

1 ____

____

2 ____

____

Do not write outside the box

0 3 . 3 Human gametes are also produced by cell division.

Where are gametes made in the human body?

[1 mark]

____

0 3 . 4 Complete the sentence.

Choose the answer from the box.

[1 mark]

doubled	**stayed the same**	**halved**

After cell division to produce gametes the

chromosome number has ______________________.

0 3 . 5 A human baby will inherit chromosomes from both of its parents.

How many **pairs** of chromosomes does a human baby have?

[1 mark]

____

0 3 . 6 A baby will inherit alleles for the ABO blood group from its parents.

What are alleles?

[1 mark]

____

____

Question 3 continues on the next page

Turn over ►

Do not write outside the box

Back pain is a long-term problem for some people.

Back pain can be treated using stem cells.

0 3 . 7 What are stem cells?

[2 marks]

0 3 . 8 Back pain can also be treated by having nerve block injections.

Figure 5 gives information about two treatments for back pain.

Figure 5

Stem cells	Nerve block injections
Stem cells are taken from the patient's bone marrow using a needle. The stem cells are injected into the correct area using guidance from X-ray images.	Anaesthetic and other medication is injected into the correct area using guidance from X-ray images or ultrasound (sound wave images).
The stem cells encourage damaged tissues to repair themselves.	Treats the symptoms but does not remove the source of pain.
Repair takes many weeks to occur.	Pain relief may be instant.
Repeat treatment should **not** be required.	Injections need to be repeated three times each year.
80% of patients experience less or no pain after treatment.	46% of patients report significant reduction in pain after treatment.
Risks • mild soreness is common • infection is rare.	Risks (all rare) • infection • bleeding • allergic reactions • spinal cord damage.
Used to treat back pain since 2010.	Used to treat back pain since before 1980.

Do not write outside the box

Evaluate the use of stem cells rather than using nerve block injections for treating back pain.

[6 marks]

15

Turn over for the next question

Turn over ►

Do not write outside the box

0 4 This question is about cells, tissues and organs.

0 4 . 1 Complete the sentence.

[2 marks]

A tissue is a group of cells with a similar ____________________

and ____________________.

0 4 . 2 What are organs made up of?

[1 mark]

Tick (✓) **one** box.

Bacteria ☐

Plasma ☐

Tissues ☐

0 4 . 3 What is a group of organs that carry out a particular function called?

[1 mark]

Tick (✓) **one** box.

An organ mass ☐

An organ network ☐

An organ system ☐

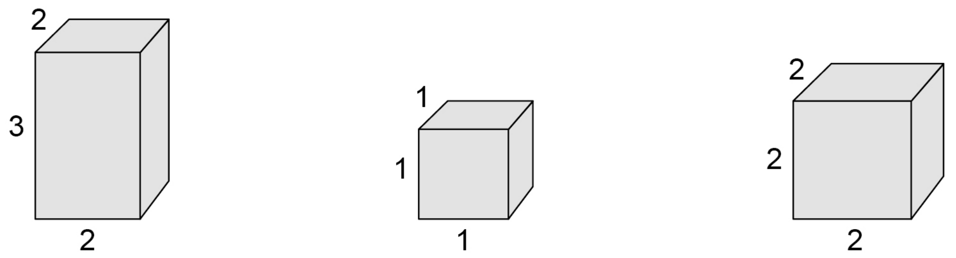
Single-celled organisms exchange gases across their outer surface membranes.

Figure 6 shows models of three single-celled organisms.

All measurements are in millimetres (mm).
SA : Vol = surface area to volume ratio

Figure 6

Organism A | **Organism B** | **Organism C**



Organism A: Surface area = 32 mm^2
Volume = 12 mm^3
SA : Vol = 2.7 : 1

Organism B: Surface area = 6 mm^2
Volume = 1 mm^3
SA : Vol = 6 : 1

Organism C: Surface area = 24 mm^2
Volume = 8 mm^3

0 4 . 4 What is the SA : Vol of **Organism C**?

[1 mark]

Tick (✓) **one** box.

1 : 1 ☐ 2 : 1 ☐ 3 : 1 ☐

0 4 . 5 The models in **Figure 6** represent organisms living in the same pond.

Which organism will have the fastest exchange rate of gases with the pond water?

[1 mark]

Tick (✓) **one** box.

Organism A ☐

Organism B ☐

Organism C ☐

Question 4 continues on the next page

One organ in the human body is the heart.

The heart pumps blood around the body.

The heart has valves.

0 4 . 6 What is the function of the valves in the heart?

[1 mark]

0 4 . 7 In some people valves may become faulty.

Describe the **two** main faults of heart valves.

[2 marks]

1

2

Do not write outside the box

0 4 . 8 Faulty heart valves can be replaced by biological heart valves or by mechanical heart valves.

What are the different types of replacement heart valves made of?

[1 mark]

Biological heart valves are made of ______________________________________.

Mechanical heart valves are made of ______________________________________.

0 4 . 9 While waiting for a valve replacement a patient may need a device to allow the heart to rest.

Name the device that can be used to allow the heart to rest.

[1 mark]

____

11

Turn over for the next question

Do not write outside the box

0 5 The common cuckoo bird has unusual breeding behaviour.

0 5 . 1 The male common cuckoo attracts a female with two signals.

Complete the sentences.

Choose answers from the box.

[2 marks]

chemical	**sound**	**visual**

Typical 'cuckoo' calls are ________________________ signals.

Dancing with drooped wings is a ________________________ signal.

Figure 7 describes the breeding behaviour of the common cuckoo.

Figure 7

Mating A male attracts a female and the pair mate for life.
Egg-laying The female lays up to 25 eggs in one season. Cuckoo birds do not build a nest of their own. The female lays one egg in each nest of a particular species of host bird. The female then removes one of the host bird's eggs and destroys it. The eggs of cuckoos look like the eggs of the chosen host species. The egg is laid very quickly while the host female is distracted.
Hatched chicks The cuckoo egg hatches in 12 days: earlier than the host bird's eggs. The hatched cuckoo chick is blind and has a loud begging call for food from the host parents. The cuckoo chick grows quickly. It will push host eggs and chicks out of the nest.

Do not write outside the box

0 5 . 2 What type of behaviour is pushing host eggs or chicks from the nest?

[1 mark]

Tick (✓) **one** box.

Classic conditioning ☐

Habituation ☐

Imprinting ☐

Innate ☐

0 5 . 3 Describe how **three** of the behaviours of the common cuckoo ensure the survival of the species.

Do **not** refer to pushing host eggs or chicks from the nest.

[3 marks]

1 ____________________

____

2 ____________________

____

3 ____________________

____

0 5 . 4 Explain the advantage for the cuckoo of having only one mate for life.

[3 marks]

____

____

____

____

____

____

9

0 6 All cells respire.

0 6 . 1 Complete the balanced **symbol** equation to represent aerobic respiration.

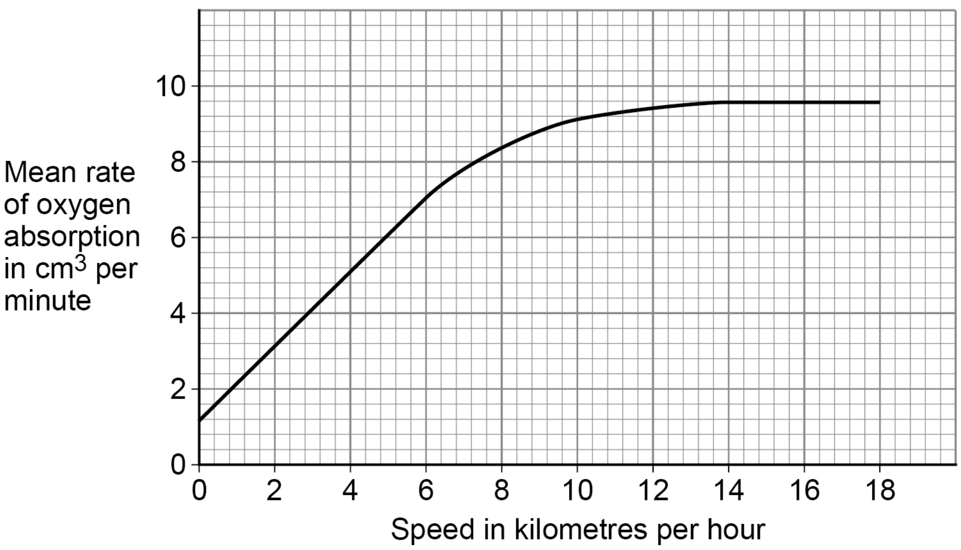
[2 marks]

__________ + __________ → __________ + __________

Students investigated the effect of running speed on the volume of oxygen absorbed.

Figure 8 shows the mean rate of oxygen absorption by the lungs at different running speeds.

Figure 8



0 6 . 2 Describe the trends shown in **Figure 8**.

Use data in your answer.

[3 marks]

0 6 . 3 Explain why the heart rates of the students increased when they started running.

[4 marks]

Question 6 continues on the next page

Do not write outside the box

0 6 . 4 Explain why many of the students developed muscle fatigue when running at 16 kilometres per hour.

[3 marks]

0 6 . 5 After the students stopped running at 16 kilometres per hour, their heart rates remained high for several minutes.

Explain why the heart rate of the students remained high.

[2 marks]

0 6 . 6 Yeast is a microorganism that can live in an environment with very little oxygen.

Which **two** products are formed when yeast respires without oxygen?

[1 mark]

1

2

15

0 7 Bacteria, viruses and fungi are microorganisms.

Some microorganisms cause infectious diseases.

0 7 . 1 What is the scientific term for microorganisms that cause infectious diseases?

[1 mark]

____

0 7 . 2 Antibiotics kill bacteria but do not kill viruses.

Why is it difficult to develop drugs to kill viruses?

[1 mark]

____

____

0 7 . 3 Explain why it is important to develop new antibiotics to kill infectious bacteria.

[3 marks]

____

____

____

____

____

____

Question 7 continues on the next page

A drug company is developing a new antibiotic, **P**, to treat infections.

The scientists at the drug company carried out three investigations.

The **first** investigation was on the diffusion of antibiotic **P** through agar gel.

This is the method used.

1. Prepare a plate of agar gel in a Petri dish.
2. Soak a paper disc in antibiotic **P** and place in the centre of the agar gel.
3. Wait 24 hours for antibiotic **P** to diffuse outwards from the disc into the agar gel.
4. Take samples of the agar gel at various distances from the disc.
5. Find the concentration of antibiotic **P** in each sample.

Figure 9 shows how the apparatus was set up.

Figure 9

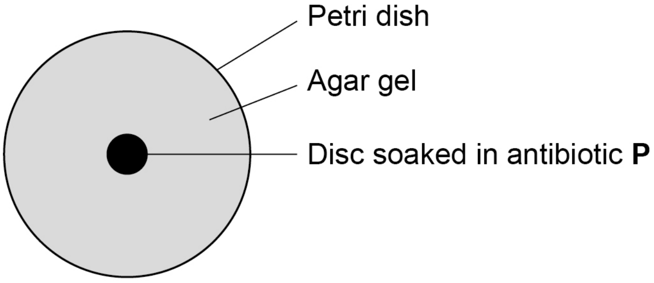


Table 2 shows the results of the **first** investigation.

Table 2

Sample	Distance from disc in centimetres	Concentration of antibiotic P in arbitrary units
1	0.10	10.0
2	0.20	5.0
3	**X**	4.0
4	0.33	3.0
5	0.50	2.0
6	1.00	1.0
7	1.43	0.7

Do not write outside the box

The scientists found that the concentration of antibiotic **P** varied according to the equation:

$$\text{concentration of antibiotic} = \frac{1}{\text{distance from the disc}}$$

0 7 . 4 **X** is the distance from the disc where the concentration of antibiotic **P** was 4.0 arbitrary units.

Calculate value **X** in **Table 2**.

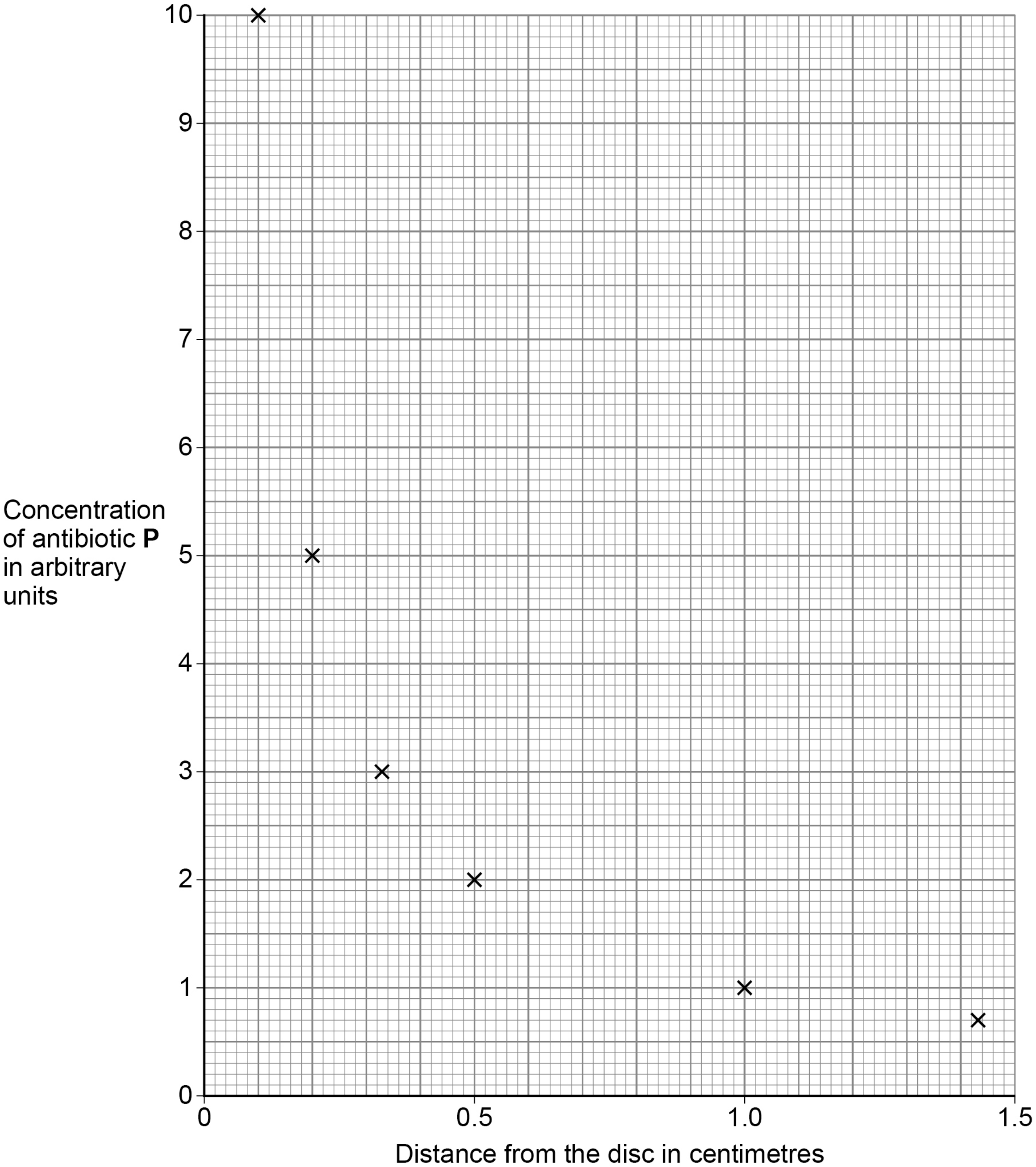
[3 marks]

X = ______________________ cm

Question 7 continues on the next page

Figure 10 shows the results of the **first** investigation.

Figure 10



0 7 . 5 Name the type of mathematical relationship shown in **Table 2** on page 24 and **Figure 10**.

[1 mark]

____

Do not write outside the box

In the **second** investigation, the scientists added bacteria to an agar gel plate before placing a disc soaked in antibiotic **P** on the plate.

The scientists found that:

- all the bacteria were killed by antibiotic **P** at concentrations above 1.5 arbitrary units
- a clear circular zone appeared in 24 hours where there were no bacteria.

0 7 . 6 Calculate the area of the clear zone.

Use **Figure 10**.

Use π = 3.14

Give your answer to 3 significant figures.

[5 marks]

Area of clear zone (3 significant figures) = ____________________ cm^2

Question 7 continues on the next page

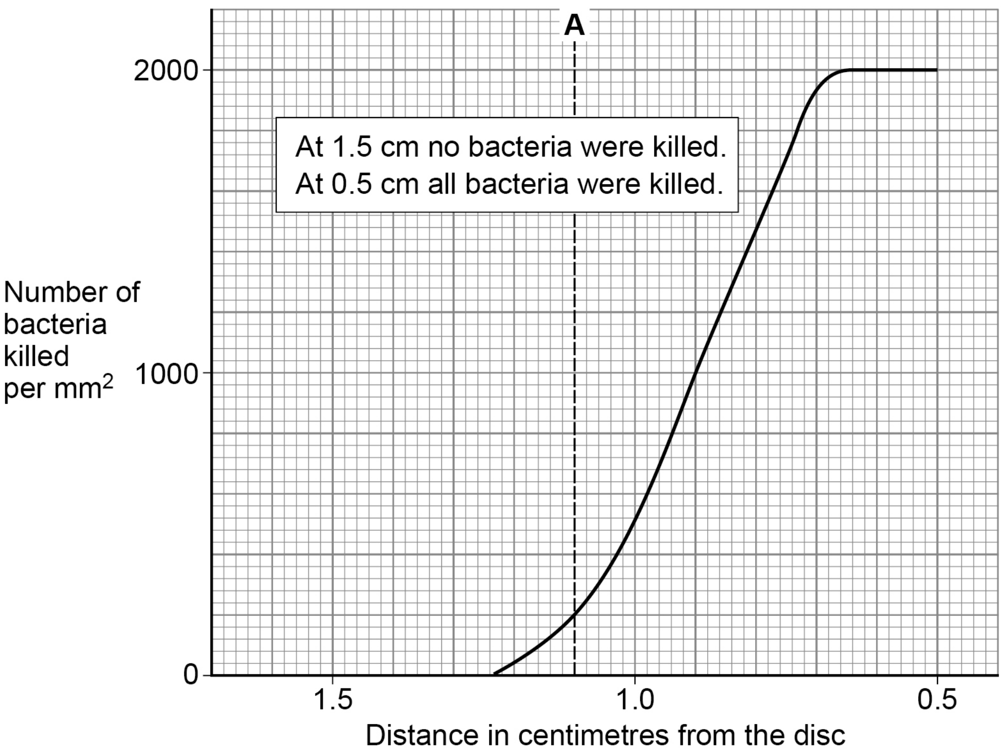
The scientists carried out a **third** investigation.

This is the method used.

1. Apply the bacteria to another agar gel plate so that the density is 2 000 bacteria per mm^2.
2. Place a disc soaked in antibiotic **P** in the centre of the agar gel plate.
3. Incubate for 24 hours.
4. Take samples at 0.2 cm distances from the disc.
5. Count the number of bacteria present per mm^2.
6. Calculate the number of bacteria killed.

Figure 11 shows the results of the **third** investigation.

Figure 11



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0 7 . 7 Line **A** on **Figure 11** shows the first decile for this data.

Describe what line **A** tells you about this data.

[1 mark]

0 7 . 8 Draw a line on **Figure 11** to show the 50th percentile for this data.

[1 mark]

16

END OF QUESTIONS

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ANSWER IN THE SPACES PROVIDED

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Question number	**Additional page, if required. Write the question numbers in the left-hand margin.**

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